Heterogeneity-Based Survival Analysis of NBI Condition Ratings for Concrete Highway Bridge Decks in Oregon by Condition Group

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Abstract: This study applied a heterogeneity-based survival analysis to determine significant contributing factors to a bridge deck's time-in-condition rating (TICR) by condition rating (CR) group: high CR, medium CR, and low CR. Although previous work has shown the effects of different CR groups on time in condition, CR-group-specific effects of NBI-related variables on time in condition by CR group are less documented. Additionally, heterogeneity issues with NBI data have focused on between-group heterogeneity, and not on bridge-deck-specific variation due to unobserved heterogeneity. In the context of statistical and econometric models, unobserved heterogeneity refers to unobserved characteristics that may influence the process of interest—in this case, the likelihood of being assigned a lower CR. This work uniquely addressed both these gaps. First, three separate survival models by CR group were fit and a parameter transferability test was conducted. Results from the parameter transferability test indicated that separate models by CR group should be fit (i.e., parameters and their effects are not transferable among CR groups). Secondly, to account for bridge-deck-specific variation due to unobserved heterogeneity and to address potential bias in parameter estimates, random parameters were estimated. Model specifications suggested that the NBI data are susceptible to large amounts of heterogeneity, as indicated by the number of random parameters and the heterogeneity-based model being preferred over the fixed parameters model. For CR-group-specific variables and effects, only two variables were found to be significant in each CR group: mountainous climate, and design period after 1970. For both variables, the effects differed by CR group. **DOI: 10.1061/(ASCE)IS.1943-555X.0000636.**

Introduction

Researchers have developed several families of statistical and probabilistic approaches to understand factors that affect the deterioration of bridge components. A popular probabilistic approach involves the use of Markovian models. In Markov models, the conditional probability of a bridge being in any future state is dependent only on the present state, and is independent of its history. A transition probability matrix and the probability distribution of the initial states describe the system evolution (Kulkarni 2016). Scherer and Glagola (1994) recommended the use of the counting process and hierarchical grouping of bridges with common characteristics to tackle the issue of the exponential increase in system states or the state space explosion issue. Poisson distributions were found to be appropriate in modeling the transition probabilities for bridges in Virginia. Jiang et al. (1988), Setunge and Hasan (2011), Morcous and Hatami (2011), and Hatami and Morcous (2012) used relative frequencies to estimate the transition probabilities. Jiang et al. (1988) focused on select state-owned bridges in Indiana; Setunge and Hasan (2011) used data from two local councils in Victoria, Australia; and Morcous and Hatami (2011) and Hatami and Morcous (2012) analyzed the Nebraska National Bridge Inventory (NBI) database. DeStefano and Grivas (1998) applied the nonparametric Kaplan–Meier method to determine transition probabilities and uniform distributions to model state transition times of 123 bridges in southern New York. Morcous et al. (2003) used a genetic algorithm to estimate the optimal combination of factors—highway class, region, average daily traffic, and percentage of truck levels—associated with each environmental category for bridges in Quebec, Canada.

Other researchers used different regression approaches to estimate transition probabilities. The regression approaches made it possible to capture the effects of environmental, construction, and traffic-related variables. Madanat et al. (1995) determined the transition probability using an ordered probit regression approach in which the dependent variable was the difference in observed condition rating in two inspections. The model was calibrated on the NBI data for the state of Indiana. Madanat and Ibrahim (1995) developed a negative binomial regression model to determine the transition probabilities. The dependent variable chosen was the number of decreases in condition rating in an inspection period. Madanat et al. (1997) further improved Madanat et al.'s (1995) probit model by incorporating a random effect in the error term to capture the unmeasured heterogeneity across various bridges. Furthermore, state dependence, i.e., the dependence of future deterioration on past states, was captured using a lagged dependent variable. Agrawal et al. (2010) compared Weibull regression with Markov chain approaches using condition rating data of bridges in the state of New York. Weibull regression was found to be more reliable in predicting deterioration than were Markov chain approaches.

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However, Jiang (2010) found Markov chain to be better at predicting deterioration than polynomial regression for bridges in the Indiana bridge database system.

Within regression approaches, duration models based on survival analysis techniques increasingly are being used to model bridge deterioration. They are more suited to model time to event or failure time, and can account for censored data (Hosmer et al. 2008). Duration models vary based on the shape assumed for the baseline hazard function. Mauch and Madanat (2001), Goyal et al. (2017), Fang et al. (2018), Wettach-Glosser et al. (2019), Goyal et al. (2020), and Wettach-Glosser et al. (2020) used a semiparametric Cox regression approach in which no specific form was assumed for the baseline hazard function shape. Goyal et al. (2020,2017) calibrated their models on the state of North Carolina bridge inventory data. Mauch and Madanat (2001), Wettach-Glosser et al. (2020), and Fang et al. (2018) developed their models using Indiana NBI, Oregon NBI, and bridge inspection data from East Japan and Tokyo, respectively. Whereas Goyal et al. (2017, 2020) used a multiple subset variable selection procedure, Wettach-Glosser et al. (2020) adopted a stepwise and least absolute shrinkage and selection operator (LASSO) for selecting variables. Other researchers followed a parametric approach in which the survival times followed a specific probability density function (PDF). Mishalani and Madanat (2002) adopted a Weibull PDF for survival time for the state of Indiana NBI database. A Weibull PDF also was found to best model the bridge deterioration in the Florida NBI (Sobanjo et al. 2010), Pennsylvania bridge management (Manafpour et al. 2018), and national NBI database (Fleischhacker et al. 2020). Tabatabai et al. (2011) found a hypertabastic PDF to be a better fit than Weibull, log logistic, and lognormal distributions for the Wisconsin NBI database. Mishalani and Madanat (2002), Sobanjo et al. (2010), and Fleischhacker et al. (2020) followed the proportional hazard assumption, whereas Tabatabai et al. (2011) and Manafpour et al. (2018) used the accelerated failure time (AFT) approach. In the proportional hazard model, the ratio of hazards of two different observations are time-independent and the effect of variables on the hazard function is modeled. In the AFT approach, the variables are assumed to have a multiplicative relationship with the survival time (Hosmer et al. 2008). Mishalani and Madanat (2002), Manafpour et al. (2018), and Goyal et al. (2020) used the final calibrated model to estimate further the transition probabilities, whereas other works focused on studying the impact of predictors on hazard or survival functions. Yang et al. (2013) adopted a mixed Weibull distribution approach which combined the advantages of both semiparametric and parametric approaches to study the deterioration of expansion joints of bridges in Hong Kong.

Artificial intelligence–based approaches also have been used to predict bridge deterioration. Morcous et al. (2002) used a case-based reasoning (CBR) approach to predict future deterioration of bridges in Quebec, Canada. In CBR, the future condition of a bridge is estimated by studying the deterioration pattern of a similar bridge. Huang (2010) applied a multilayer artificial neural network model to predict deterioration of bridge decks in Wisconsin based on 11 factors, including maintenance history, age, traffic, environmental, and design variables. Radovic et al. (2017) and Galvan-Nunez and Attoh-Okine (2017) adopted clustering-based analysis to identify key variables associated with structural deficiencies and deterioration in the northeast US and national NBI databases, respectively. Whereras Radovic et al. (2017) used a two-step cluster analysis approach, Galvan-Nunez and Attoh-Okine (2017) focused on using a particle swarm optimization metaheuristic to optimize k-means clustering objectives.

Bolukbasi et al. (2004) determined deterioration curves of each CR for deck, superstructure, and substructure based on cumulative durations utilizing the Illinois NBI database. Kim and Yoon (2010) applied linear regression to study the impact of physical, material, environmental, maintenance, and repair factors on the deterioration of bridges using the North Dakota NBI database. Chang et al. (2017) used covariance analysis and linear regression based LASSO framework to identify key explanatory variables. These explanatory variables were used to develop deterministic deterioration models for the Wyoming NBI database. Chang et al. (2019) further developed this work by categorizing the bridges into subsets using classification trees based on key variables identified using LASSO. A Markov chain model was built to predict deterioration, in which the transition probabilities were estimated using a zoning method as well as logistic regression. Hasan and Elwakil (2020) developed separate linear regression models for each bridge service to predict the deck condition of California NBI bridges based on geometric and design variables other than age and maintenance. Ghonima et al. (2020) developed a random parameter–based binary logistic regression model to understand the impact of various structural, environmental, traffic and maintenance-related variables on the likelihood of a bridge deck being associated with highest and lowest deterioration rates. The random-parameter model was used to capture heterogeneity in a select subset of bridges in the nationwide NBI database.

Although all the aforementioned models are frequentist in nature, with the exception of that of Fleischhacker et al. (2020), researchers also have used Bayesian approaches. Bulusu and Sinha (1997) compared Bayesian approaches for updating transition probability with binary probit regression models with state dependence accounted for using a lagged state transition term and heterogeneity captured using a bridge-specific error term. The models were calibrated on the Indiana NBI database. No statistical differences were found in the predictions of the simpler Bayesian approach and the binary probit regression approach. Morcous (2006) applied Bayes' rule to adjust the transition probabilities in Markovian models to account for inspection period variation. Wellalage et al. (2015) developed a Metropolis Hasting algorithm–based Markov chain Monte Carlo (MCMC) simulation technique. This algorithm was better at estimating transition probabilities than were regression-based nonlinear optimization and the Bayesian maximum likelihood approach for Australian railway bridges. Mishalani et al. (2018) found Bayesian updating to be superior to maximum likelihood in estimating transition probabilities for bridges in the Ohio Department of Transportation bridge inspection database. Li and Jia (2020) used a Bayesian framework to update the probability density functions of time in condition rating and emphasized the value of considering censored data. The analysis was conducted on the Colorado NBI database. Fleischhacker et al. (2020) used the Hamiltonian Monte Carlo algorithm to estimate a Weibull-based proportional hazard model for the nationwide NBI database.

Several researchers identified heterogeneity as an issue with bridge inspection data and models due to factors such as unobserved data (Madanat et al. 1997; Bulusu and Sinha 1997; Ghonima et al. 2020). Whereas Madanat et al. (1997) and Bulusu and Sinha (1997) captured heterogeneity using a random effects probit model in a panel framework, Ghonima et al. (2020) adopted a random-parameters logistic regression approach. These three methodological approaches, however, do not appropriately account for the inherent censorship of such data. To adequately account for this censorship, a survival-based analysis can be applied. As previously outlined, various survival-based studies have been conducted using NBI data (Goyal et al. 2020; Wettach-Glosser et al. 2020; Manafpour et al. 2018; Goyal et al. 2017; Tabatabai et al. 2011; Mishalani and Madanat 2002; Mauch and Madanat 2001); however, analyses focusing explicitly on the impacts of unobserved heterogeneity (unobserved factors that may influence the outcome of interest) on parameter estimates are scarce. Additionally, although analyses have been done by CR group, no parameter transferability test has been conducted to determine, statistically, if separate models should be fit contingent on CR group. This work addressed these gaps by considering data heterogeneity through the estimation of random parameters in a survival regression framework. For the transferability test, this work determined if estimated parameters by CR group are transferable (i.e., if separate models should be fit by CR group).

Data

The data used for the current study were a subset of the National Bridge Inventory database (Federal Highway Administration 2020); this work focused on concrete highway bridge decks in Oregon. Two additional variables, climate zone and design period, were created based on information in the NBI data; Wettach-Glosser et al. (2020, 2019) gave full details.

A key performance metric of interest was a bridge deck's time-in-condition rating (TICR) (Nasrollahi and Washer 2015). TICR, as illustrated in the methodology, is the premise behind the survival analysis. TICR is defined as the number of years a bridge deck is in the same NBI condition rating (CR). Multiple TICR occurrences may be observed over the duration of available CR records, which for the current study ranged from 1992 to 2016. Because CRs are available only for a limited number of years relative to the life of a bridge deck, censorship takes place. This censorship occurs when the value of TICR, as derived from observed condition ratings, is only partially observable (Greene 2018; Wooldridge 2010). For example, for a TICR to be considered uncensored, both the start and end have to be observed with the condition that the subsequent CR value decreases. The case in which the subsequent CR value increases is considered censored, because it implies maintenance action that interferes with a deck's natural deterioration process. Cases in which either the start or end, or both, cannot be observed, or in which more than three CR values are missing, impose censorship. For all cases, censorship signifies that TICR is a lower bound value that may be longer. On this premise, this study did not consider interval-based censorship, but only right-censorship.

This study focused on contributing factors to TICR by CR group, so the data were disaggregated into three specific CR groups. The three groups were based on findings from Wettach-Glosser et al. (2019), in which hazard ratios for CR 7 and CR 8 were similar, hazard ratios for CR 4 and CR 5 were similiar, and hazard ratios for CR 6 did not follow the trends of either. Therefore, the three CR groups considered in this work were as follows:

- high CR: CR rating of 7 or 8;
- medium CR: CR rating of 6; and
- low CR: CR rating of 4 or 5.

Before disaggregating, the holistic data set consisted of 11,940 observations of bridge decks with CRs ranging from 3 to 9. Tables 1–3 summarize the statistics of significant variables by CR group. Each table provides the associated NBI item number, if applicable, as defined in the NBI recording and coding guide (Federal Highway Administration 1995). In the high-CR group (Table 1), approximately 51% of bridges were under the maintenance responsibility of state highway (state responsible) (NBI Item 21-1). In terms of the kind of material design (NBI Item 43A), 46.7% were prestressed concrete (43A-5), 23.2% were continuous concrete (43A-2), and 8.7% were steel (43A-3). In terms of the type of design construction (NBI Item 43B), 34.5% were slab design (43B-1), and 8.8% were box beam or girders (43B-5). In terms of the the type of wearing surface (NBI Item 108A), 25.1% had a monolithic concrete wearing surface (108A-1), and 3% had an epoxy overlay wearing surface (108A-5). In the high-CR group, 87.9% of bridge decks did not have a membrane (NBI Item 108B-1). The final variable linked to an NBI item was average daily truck traffic (ADTT). Based on natural breaks in the distribution of ADTT on Oregon bridges, an indicator for high ADTT (greater than 1,000) was created (ADTT is NBI Item 109). In the high-CR group, 22.5% of bridges had an ADTT of greater than 1,000.

Considering the medium-CR group, roughly 74% fell under the maintenance responsibility of state highway (state responsible) and 24% under the maintenance responsibility of county highway (county responsible) (NBI Item 21-2). In terms of the kind of material design, 38.1% were continuous concrete, 19% were steel, and 3.7% were continuous steel (NBI Item 43A-4). Deck structure type was found to be significant only in the medium-CR group, in which 91.9% were cast-in-place (NBI Item 107-1). The only significant variable in the medium-CR model was type of design construction, in which 55.4% were stringer/multibeam girders (NBI Item 43B-2). In terms of the type of wearing surface (NBI Item 108A), 3.9% had no wearing surface (108A-0), 81.2% had an epoxy overlay wearing surface (108A-5), and 42.5% had a bituminous wearing surface (108A-6). In terms of ADTT, 34.9% of bridges in the medium-CR group had ADTT greater than 1,000.

In the final CR group, low CR of 4 and 5, only six variables were found to be significant. In this group, 21.1% fell under the maintenance responsibility of county highway (county responsible) (NBI Item 21-2), 7.3% had no wearing surface (NBI Item 108A-0), 18.8% had a latex concrete wearing surface (NBI Item 108A-3), and 5.5% had a type of design consisting of a through truss (NBI Item 43B-10).

Additionally, the number of observations substantially decreased from the high-CR group to the low-CR group (Tables 1–3). Specifically, the high-CR group had a total of 8,309 observations, of which 6,375 were censored (76.7%); the medium-CR group had 2,699 observations, of which 2,201 were censored (81.5%); and the low-CR group had 1,045 observations, of which 944 were censored (90.3%). The summary statistics in Tables 1–3 reflect the statistics of the data used for analysis, not statistics based on the results of the analysis.

Methodology

Random Parameters Survival Model

As discussed previously, by its nature TICR experiences a relatively high percentage of right-censoring, resulting in the need for a specific modeling framework that can account for such censorship. This study implemented a survival-based model to account for the censored nature of TICR. A commonly used approach is the semiparametric Box–Cox proportional hazards model. However, the use of the Box–Cox proportional hazards model (i.e., a semiparametric model) does not allow the analyst to account for data heterogeneity, only for heteroscedasticity (Greene 2016). Therefore, upon assessing various parametric survival models under different distributions (a parametric model needs to be considered to account for bridge-deck-specific variation through random parameters estimation), it was determined based on the estimated integrated hazard function that the best-fit distribution for the parametric survival models of TICR was Weibull. This agrees with previous research which found that the Weibull distribution is well suited for the application of TICR (Nasrollahi and Washer 2015; Fleischhacker et al. 2020). Most notably, the Weibull distribution can model a failure rate, or TICR, that increases or decreases with time (a shape parameter greater than 1 results in an increasing rate over time, and a shape

Table 1. Summary statistics of significant variables in high-CR (7 and 8) model

Variable	Mean (number of observations)[a]	Standard deviation
Dependent variable		
Time-in-condition rating	7.570	6.054
Maintenance responsibility		
1 if state highway (state responsible), 0 otherwise (NBI item 21-1)	0.510 (4,103)	0.500
Kind of material design		
1 if concrete continuous, 0 otherwise (NBI item 43A-2)	0.232 (1,864)	0.422
1 if steel, 0 otherwise (NBI item 43A-03)	0.087 (699)	0.282
1 if prestressed concrete, 0 otherwise (NBI Item 43A-5)	0.467 (3,757)	0.500
Type of design construction		
1 if slab, 0 otherwise (NBI item 43B-1)	0.345 (2,776)	0.476
1 if box beam or girder, 0 otherwise (NBI Item 43B-5)	0.088 (3,227)	0.283
Type of wearing surface		
1 if monolithic concrete, 0 otherwise (NBI item 108A-1)	0.251 (2,017)	0.434
1 if epoxy overlay, 0 otherwise (NBI item 108A-5)	0.030 (245)	0.172
Type of membrane		
1 if no membrane, 0 otherwise (NBI Item 108B-0)	0.879 (7,070)	0.326
Deck protection		
1 if no deck protection, 0 otherwise (NBI item 108C-0)	0.847 (6,809)	0.326
Average daily truck traffic		
1 if greater than 1,000, 0 otherwise (NBI item 109)	0.225 (1,789)	0.416
Distance to seawater		
1 if less than or equal to 1 km, 0 otherwise	0.014 (111)	0.117
Climate group		
1 if Climate group 2, 0 otherwise[b]	0.153 (1,229)	0.360
Design period		
1 if before 1950, 0 otherwise	0.073 (584)	0.260
1 if after 1970, 0 otherwise	0.515 (4,141)	0.500

[a]Total number of observations = 8,039.
[b]Climate group 2 includes Cascade Mountain Range.

parameter less than 1 results in a decreasing rate over time) (Fleischhacker et al. 2020).

In terms of survival models, it was assumed that random variable T (TICR) has a continuous probability distribution, denoted $f(t)$. Therefore, the cumulative probability of TICR, $F(t)$, can be defined as (Greene 2018; Wooldridge 2010)

$$F(t) = \int_0^t f(s)ds = P(T \leq t) \tag{1}$$

where the interest, generally, is in the probability that TICR is of length of at least t, given by the following survival function, $S(t)$ (Greene 2018; Wooldridge 2010):

$$S(t) = 1 - F(t) = P(T \geq t) \tag{2}$$

Using the cumulative probability and survival function, the probability that TICR will end (i.e., a bridge deck will be assigned a lower CR) by the next interval of time (intervals of 1 year were used in this work), Δt, can be represented as

$$P(t \leq T \leq t + \Delta t | T \geq 1) \tag{3}$$

The hazard rate, $\lambda(t)$, is used to characterize this as the instantaneous variant of the preceding probability (Greene 2018; Wooldridge 2010)

$$\lambda(t) = \lim_{\Delta t \to 0} \frac{P(t \leq T \leq t + \Delta t | T \geq t)}{\Delta t} = \lim_{\Delta t \to 0} \frac{F(t + \Delta t) - F(t)}{\Delta t S(t)} = \frac{f(t)}{S(t)} \tag{4}$$

where $f(t)/S(t)$ = hazard rate, defined as the rate at which TICR has changed after duration t, given that TICR has lasted until t. Under these assumptions, because the density, cumulative probability, and hazard function are all related, the hazard function also can be represented as (Greene 2018; Wooldridge 2010)

$$\lambda(t) = \frac{-d \ln[S(t)]}{dt} \tag{5}$$

Of further interest is the integrated hazard function, which is used to determine the appropriate distribution for the parametric survival model (Greene 2016). The integrated hazard function is (Greene 2018; Wooldridge 2010)

$$\Delta t = \int_0^t \lambda(s)ds \tag{6}$$

Using the presented formulas, the density, survival, and hazard functions can be expressed in terms of the Weibull distribution (the distribution used for survival analysis in this study). The density, survival, and hazard functions, respectively, under the Weibull distribution are (Greene 2016)

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Table 2. Summary statistics of significant variables in medium CR (6) model

Variable	Mean (number of observations)[a]	Standard deviation
Dependent variable		
Time-in-condition rating	6.657	5.053
Maintenance responsibility		
1 if state highway (state responsible), 0 otherwise (NBI Item 21-1)	0.739 (1,994)	0.439
1 if county highway (county responsible), 0 otherwise (NBI item 21-2)	0.239 (645)	0.427
Kind of material design		
1 if concrete continuous, 0 otherwise (NBI item 43A-2)	0.381 (1,029)	0.486
1 if steel, 0 otherwise (NBI Item 43A-3)	0.190 (514)	0.393
1 if steel continuous, 0 otherwise (NBI item 43A-4)	0.037 (101)	0.190
Deck structure type		
1 if cast-in-place, 0 otherwise (NBI item 107-1)	0.919 (2,481)	0.273
Type of design construction		
1 if stringer/multibeam girder, 0 otherwise (NBI Item 43B-2)	0.554 (1,494)	0.497
Type of wearing surface		
1 if no wearing surface, 0 otherwise (NBI Item 108A-0)	0.039 (104)	0.193
1 if epoxy overlay, 0 otherwise (NBI item 108A-5)	0.812 (220)	0.274
1 if bituminous, 0 otherwise (NBI item 108A-6)	0.425 (1,146)	0.494
Average daily truck traffic		
1 if greater than 1,000, 0 otherwise (NBI Item 109)	0.349 (942)	0.477
Climate group		
1 if Climate group 2, 0 otherwise[b]	0.216 (583)	0.412
Design period		
1 if after 1970, 0 otherwise	0.327 (882)	0.469

[a]Total number of observations = 2,699.
[b]Climate group 2 includes Cascade Mountain Range.

Table 3. Summary statistics of significant variables in low CR (4 and 5) model

Variable	Mean (number of observations)[a]	Standard deviation
Dependent variable		
Time-in-condition rating	5.667	4.417
Maintenance responsibility		
1 if county highway (county responsible), 0 otherwise (NBI item 21-2)	0.211 (221)	0.409
Type of wearing surface		
1 if no wearing surface, 0 otherwise (NBI item 108A-0)	0.073 (76)	0.260
1 if latex concrete, 0 otherwise (NBI item 108A-3)	0.188 (196)	0.391
Type of design construction		
1 if through truss, 0 otherwise (NBI item 43B-10)	0.055 (57)	0.227
Climate group		
1 if Climate group 2, 0 otherwise[b]	0.243 (254)	0.429
Design period		
1 if after 1970, 0 otherwise	0.240 (251)	0.427

[a]Total number of observations = 1,045.
[b]Climate group 2 includes Cascade Mountain Range.

$$f(t) = \lambda p(\lambda t)^{p-1} \cdot e^{[-(\lambda t)^p]} \tag{7}$$

$$S(t) = e^{[-(\lambda t)^p]} \tag{8}$$

$$h(t) = \lambda p(\lambda t)^{p-1} \tag{9}$$

For the Weibull survival model, model estimation relies on the following transformation, creating a log-linear survival model (Greene 2016):

$$w = \frac{\log(t - \boldsymbol{\beta})}{\sigma} \tag{10}$$

where

$$\lambda = e^{-\boldsymbol{\beta}} \tag{11}$$

$$p = \frac{1}{\sigma} \tag{12}$$

After transforming, the density function and survival function for the now log-linear Weibull survival model are, respectively (Greene 2016)

$$f(t) = e^{\{[\log(t-\boldsymbol{\beta})]/\sigma\} - e^{[\log(t-\boldsymbol{\beta})]/\sigma}} \tag{13}$$

$$S(t) = e^{-e^{[\log(t-\boldsymbol{\beta})]/\sigma}} \tag{14}$$

where σ = scale parameter for Weibull distribution. To estimate parameters for this log-linear Weibull survival model considering uncensored and censored observations, the likelihood function is as follows (Greene 2016):

$$\mathrm{L} = \prod_{i=1}^{n} [\sigma^{-1} f(w_i)]^{\delta_i} [S(w_i)]^{1-\delta_i} \tag{15}$$

where δ_i = censoring indicator, which equals 1 if uncensored and 0 if censored; and all other terms are as defined previously. In taking the log of both sides, the log-likelihood equation used for parameter estimation and model fit is expressed as (Greene 2016)

$$\mathrm{LL} = \sum_{i=1}^{n} [\delta_i(-\log\sigma + \log h(w_i)) + \log S(w_i)] \tag{16}$$

To account for unobserved heterogeneity, the manner in which parameters are estimated also changes. The ability to account for unobserved heterogeneity is possible only with a parametric form of a survival model. Unobserved heterogeneity arises from two common factors. The first of these factors is related to data being unavailable, which is most often a result of specific data not being collected. In the case of the NBI data, all factors that may contribute to a bridge deck being assigned a lower CR at any given time are not collected during inspections. This most notably occurs as a result of all factors not being collected or noted during inspection. The second factor is related to variation in observable variables due to unobserved characteristics. Examples of unobservable characteristics include information related to material properties at construction, construction and curing practice, and so forth. It is vital to account for unobserved heterogeneity, because not doing so can, and often does, result in biased parameter estimates (i.e., the estimated parameter is not an accurate representation of the true population parameter). Therefore, to account for this heterogeneity, this study estimated random parameters, and $\boldsymbol{\beta}$ now is estimated as (Greene 2016; Washington et al. 2011)

$$\boldsymbol{\beta}_i = \boldsymbol{\beta} + \boldsymbol{\omega}_i \tag{17}$$

where $\boldsymbol{\omega}_i$ = randomly distributed term (e.g., normally distributed with mean zero and variance σ^2). The addition of $\boldsymbol{\omega}_i$ now allows $\boldsymbol{\beta}$ to vary across observations (bridge decks), therefore accounting for bridge-deck-specific variation (unobserved heterogeneity). The distribution of $\boldsymbol{\omega}_i$ refers to the distribution in which $\boldsymbol{\beta}$ varies across observations. Although various distributions can be considered for random parameters (e.g., lognormal, triangular, uniform, and so forth), the normal distribution has been shown to be the preferred distribution in terms of statistical fit (Mannering et al. 2016). Additionally, the variation of $\boldsymbol{\beta}$ is interpreted readily based on the normal distribution. If the estimated standard deviation in the randomly distributed term is statistically significant, the parameter is considered random. If it is random, the effects of the respective variable will vary across observations based on a normal distribution; that is, there will be an increased likelihood of being assigned a lower CR at any given time for a proportion of bridge decks and a decreased likelihood of being assigned a lower CR at any given time for the remainder (or vice versa).

The final presentation is related to interpretation of model estimates, specifically for those variables that do not have estimated random parameters. This is accomplished by computing the proportional hazards and the hazard ratios. In most statistical analysis software, these often are seen as an alternative representation of model specifications. The proportional hazards are computed as (Greene 2016)

$$c(k) = -\frac{1}{\sigma}\boldsymbol{\beta}(k) \tag{18}$$

where $c(k)$ = proportional hazard for parameter k; σ = scale parameter associated with Weibull distribution; and $\boldsymbol{\beta}(k)$ = estimate of parameter k. The hazard ratio then is computed as (Greene 2016)

$$h(k) = e^{-(1/\sigma)\cdot\boldsymbol{\beta}(k)} \tag{19}$$

where $h(k)$ = hazard ratio for parameter k; and all other terms are as defined previously. For interpretation in this work, hazard ratios less than one indicate a decrease in the likelihood of being assigned a lower CR at any given time, and hazard ratios greater than one indicate an increased likelihood of being assigned a lower CR at any given time.

Model Significance and Parameter Transferability

Log-likelihood ratio tests were applied to determine the model's significance. The log-likelihood ratio test was conducted to test the null hypothesis that the fixed-parameters model is preferred over the random-parameters model (Washington et al. 2011)

$$\chi^2 = -2[\mathrm{LL}(\boldsymbol{\beta}_{\mathrm{FP}}) - \mathrm{LL}(\boldsymbol{\beta}_{\mathrm{RP}})] \tag{20}$$

where $\mathrm{LL}(\boldsymbol{\beta}_{\mathrm{FP}})$ = log-likelihood at convergence of the fixed parameters model; $\mathrm{LL}(\boldsymbol{\beta}_{\mathrm{RP}})$ = log-likelihood at convergence of random-parameters model; and χ^2 = chi-squared distributed statistic with degrees of freedom equal to number of estimated random parameters. If the ratio test [Eq. (20)] results in statistical significance, the null hypothesis is rejected and the random-parameters model is preferred.

The next assessment is related to parameter transferability. Parameter transferability refers to the transferability of parameters from model to model. In this study, parameter transferability was assessed for three CR groups: high CR (7 and 8), medium CR (6), and low CR (4 and 5). This assessment tested the null hypothesis that parameter estimates are transferable (i.e., all CR groups should be modeled holistically in a single model with estimated parameters representing the effects of all CR groups). Therefore, if the null hypothesis is rejected, parameter estimates are not transferable and separate models by CR group should be fit (i.e., estimated parameters and their effects are contingent on CR group). To determine parameter transferability, an additional log-likelihood ratio test was conducted (Washington et al. 2011)

$$\chi^2 = -2[\mathrm{LL}(\boldsymbol{\beta}_{\mathrm{CR}_{i\mathrm{CR}_j}}) - \mathrm{LL}(\boldsymbol{\beta}_{\mathrm{CR}_i})] \tag{21}$$

where $\mathrm{LL}(\boldsymbol{\beta}_{\mathrm{CR}_i})$ = log-likelihood at convergence of CR model i (e.g., high-CR model); $\mathrm{LL}(\boldsymbol{\beta}_{\mathrm{CR}_{i\mathrm{CR}_j}})$ = log-likelihood at convergence of CR model i using data from CR model j (e.g., medium-CR model); and χ^2 is as defined previously. To illustrate, the best-fit model for high CR results in $\boldsymbol{\beta}$ estimates, both a constant and those associated with X. These $\boldsymbol{\beta}$ values estimated for the high-CR model then are fixed, and the constant is used as a starting value, and the same model is run using data from the medium CR model and the low CR model; this output is $\mathrm{LL}(\boldsymbol{\beta}_{\mathrm{CR}_{i\mathrm{CR}_j}})$. The degrees of freedom used for the transferability test, based on a chi-squared distribution, are the number of estimated parameters in $\mathrm{LL}(\boldsymbol{\beta}_{\mathrm{CR}_{i\mathrm{CR}_j}})$.

Model Estimation Results

Using Eq. (20), it was determined that two of the three random-parameters CR models were statistically preferred over the fixed-parameters models with a high level of confidence. The results of the ratio tests are listed in Table 4. With a chi-squared statistic of 50.52 and nine degrees of freedom (the number of estimated random parameters), the random-parameters model was preferred for the high-CR model with well over 99% confidence. Model specifications for the high-CR model are listed in Table 5, hazard ratios for variables with fixed parameters are shown in Fig. 1(a), and the proportion of bridge decks with an estimated mean above and below zero (variables with random parameters) are shown in Fig. 1(b). With a chi-squared statistic of 28.80 and seven degrees of freedom (the number of estimated random parameters), the random-parameters model was preferred for the medium-CR model with well over 99% confidence. Model specifications for the medium-CR model are listed in Table 6, hazard ratios for variables with fixed parameters are shown in Fig. 2(a), and the proportion of bridge decks with an estimated mean above and below zero (variables with random parameters) are shown in Fig. 2(b). However, for the low-CR model, a chi-squared statistic of 5.22 and three degrees of freedom indicated that the random-parameters model was preferred with just under 80% confidence. Model specifications for the low-CR model are listed in Table 7, hazard ratios for variables with fixed parameters are shown in Fig. 3(a), and the proportion of bridge decks with an estimated mean above and below zero (variables with random parameters) are shown in Fig. 3(b). For the McFadden pseudo R^2 values in Tables 5–7, unlike ordinary least squares (OLS)-based regression in which R^2 values are expected to be high, McFadden pseudo R^2 values can be much lower and still imply adequate fit (McFadden 1973, 1977, 1981).

For the results of the parameter transferability test (i.e., model separation by CR group), chi-squared statistics and corresponding degrees of freedom are listed in Table 8. Each chi-squared statistic and the corresponding degrees of freedom indicated with well over 99% confidence that the presented CR groups should be modeled separately (Table 8). In other words, the null hypothesis that parameter estimates are transferable is rejected. Therefore, the use of three separate models by CR group is appropriate.

Discussion

Each CR is discussed separately. This is followed by a comparison of significant variables and their effects among CR groups.

Table 4. Likelihood ratio test results of random parameters models

Model	Parameters	Log-likelihood	Degrees of freedom[a]	χ^2	p-value
High-CR model					
Fixed	15	−4,464.64	—	—	—
Random	15	−4,439.38	9	50.52	0.000
Medium-CR model					
Fixed	13	−1,380.36	—	—	—
Random	13	−1,365.96	7	28.80	0.000
Low-CR model					
Fixed	6	−353.00	—	—	—
Random	6	−350.75	3	4.50	0.212

[a]Degrees of freedom are equal to the number of random parameters.

High-CR Group

For the high-CR model (CRs of 7 and 8), 15 variables were found to be significant contributing factors to TICR. Of these 15 variables, 9 had heterogeneous effects on TICR (i.e., they had estimated random parameters).

NBI Item 43A–Structure Type: Kind of Material Design
The first two variables with normally distributed random parameters were related to the kind of material design; specifically, steel and prestressed concrete. For steel, model estimates showed a mean of −0.099 and a standard deviation of 0.479. Based on the normal distribution curve, these estimates indicate that for 41.8% of observations, the estimated parameter mean was greater than zero, and for 58.2% of observations, the estimated parameter mean was less than zero. That is, 41.8% of high-CR bridge decks with steel material design are less likely to be assigned a lower CR at any given time, whereas 58.2% are more likely to be assigned a lower CR. The heterogeneous effects observed here for steel material design can be attributed to different factors, such as age, environment, and traffic conditions (Goyal 2015). For age, the heterogeneous effects may be capturing the age of bridges, because age has been shown to significantly impact deterioration (Goyal et al. 2017). For environmental characteristics, the heterogeneous nature may be associated with bridges located near water and those that are not (Kim and Yoon 2010), because the effect of water on freezing and thawing can contribute to damage. For traffic conditions, the mean time in a specific condition rating has been shown to be contingent on road classification, because mean times in condition rating are lower for interstate roadways than for noninterstate roadways (Sobanjo et al. 2010). Another factor that may result in the heterogeneous effects may be related to paint on the steel material design (Arima et al. 1999). Additionally, several previous works have shown that the number of spans can contribute to deterioration, because multiple spans have an increased rate of deterioration (Goyal et al. 2017; Busa et al. 1985; Chen 1987; Madanat et al. 1995; Madanat and Ibrahim 1995). This most notably results from expansion joints in multispan decks that are prone to maintenance issues that can result in increased deterioration (Chang and Lee 2002). Additionally, multispan bridges have negative bending regions at the top of internal supports, which can lead to increased cracking due to tension induced in the deck. This cracking also can lead to higher deterioration. If the bridge has been reconstructed, previous work has shown higher deterioration rates, which also can be attributing to the observed varying effects.

The second material design variable with a normally distributed random parameter was prestressed concrete. With a mean of 0.310 and a standard deviation of 0.359, 19.4% of high-CR bridge decks with prestressed concrete material design are more likely to be assigned a lower CR at any given time, and 80.6% are less likely to be assigned a lower CR at any given time. The heterogeneous effects of prestressed concrete also was observed in previous work. For example, Fleischhacker et al. (2020) found that prestressed concrete is less likely to be assigned a lower CR at any given time (decreased deterioration), whereas Mauch and Madanat (2001) found that prestressed concrete leads to higher deterioration rates. Goyal et al. (2017) found that prestressed concrete on secondary routes, compared with interstates, has slower deterioration. On the other hand, Abed-Al-Rahim and Johnston (1991) found the opposite, suggesting that variation in effects were attributed to design variations on roadways with lower traffic volumes. Mauch and Madanat (2001) and Mishalani and Madanat (2002) found similar results and also attributed the findings to lower design requirements on low-volume roads, as well as to maintenance standards on secondary roads. Using this same premise, Sobanjo et al. (2010) found that

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Table 5. Model specifications for high-CR (7 and 8) random-parameters survival model

Variable	Coefficient	Standard error	t-statistic	Proportional hazards	Hazard ratio
Constant	3.851	0.100	38.51	−5.757	0.003
Maintenance responsibility					
1 if state highway (state responsible), 0 otherwise	−0.266	0.041	−6.47	0.398	1.489
Kind of material design					
1 if concrete continuous, 0 otherwise	−0.101	0.042	−2.42	0.151	1.163
1 if steel, 0 otherwise	−0.099	0.058	−1.70	0.149	1.160
(Standard deviation, normally distributed)	(0.479)	(0.046)	(10.29)	—	—
1 if prestressed concrete, 0 otherwise	0.310	0.051	6.10	−0.463	0.629
(Standard deviation, normally distributed)	(0.359)	(0.035)	(10.32)	—	—
Type of design construction					
1 if slab, 0 otherwise	0.550	0.059	9.26	−0.822	0.440
(Standard deviation, normally distributed)	(0.561)	(0.048)	(11.67)	—	—
1 if box beam or girders, 0 otherwise	0.266	0.065	4.06	−0.397	0.672
(Standard deviation, normally distributed)	(0.547)	(0.057)	(9.52)	—	—
Type of wearing surface					
1 if monolithic concrete, 0 otherwise	−0.260	0.042	−6.19	0.388	1.474
(Standard deviation, normally distributed)	(0.360)	(0.031)	(11.60)	—	—
1 if epoxy overlay, 0 otherwise	−0.223	0.082	−2.71	0.334	1.396
(Standard deviation, normally distributed)	(0.508)	(0.079)	(6.45)	—	—
Type of membrane					
1 if no membrane, 0 otherwise	−0.357	0.060	−6.00	0.534	1.706
Deck protection					
1 if no deck protection, 0 otherwise	−0.136	0.058	−2.33	0.203	1.225
Average daily truck traffic					
1 if greater than 1,000, 0 otherwise	−0.337	0.040	−8.51	0.503	1.654
Distance to seawater					
1 if less than or equal to 1 km, 0 otherwise	−0.323	0.134	−2.40	0.483	1.620
(Standard deviation, normally distributed)	(0.928)	(0.127)	(7.33)	—	—
Climate group					
1 if Climate group 2, 0 otherwise	−0.102	0.047	−2.18	0.152	1.164
(Standard deviation, normally distributed)	(0.711)	(0.041)	(17.27)		
Design period					
1 if before 1950, 0 otherwise	−0.078	0.051	−1.52	0.117	1.124
1 if after 1970, 0 otherwise	0.196	0.043	4.53	−0.294	0.745
(Standard deviation, normally distributed)	(0.369)	(0.029)	(12.77)	—	—
Scale parameter					
σ	0.669	0.016	42.86		
Model summary					
Number of observations	8,039	—	—	—	—
Number of observations exiting (uncensored)	1,644	—	—	—	—
Number of observations censored	6,375	—	—	—	—
Log-likelihood at zero	−4,791.61	—	—	—	—
Log-likelihood at convergence	−4,439.38	—	—	—	—
McFadden pseudo R^2	0.07	—	—	—	—

the mean time in a given condition rating was lower for interstate roadways than for noninterstate roadways for prestressed concrete superstructures. Other factors impacting bridge-deck-specific variation may be linked to degree skew, structure length, and deck width (Hasan and Elwakil 2020; Freyermuth et al. 1970).

NBI Item 43B–Structure Type: Type of Design/Construction

Two random parameters were related to design construction. Slab design construction had a normally distributed random parameter with a mean of 0.550 and a standard deviation of 0.561, indicating that 16.3% of high-CR bridge decks with a slab design are more likely to be assigned a lower CR at any given time and 83.7% are less likely to be assigned a lower CR at any given time. Heterogeneous effects of slab design construction may be related to the bridge panel, because panels in different positions have been shown to result in varying deterioration rates (Fang et al. 2018). Fang et al. (2018) also found that winter precipitation and deicing use can increase deterioration, both of which can vary. Variation also may be be a result of the amount of waterproofing that specific slabs have on the edge, in addition to the slab having a straight or sloped edge (Fang et al. 2018). Other work found that the level of corrosion can result in increased deterioration (Morcous et al. 2010; Kim and Yoon 2010), but level of corrosion is unobservable in the NBI data. Setunge and Hasan (2011) found that slab design construction has

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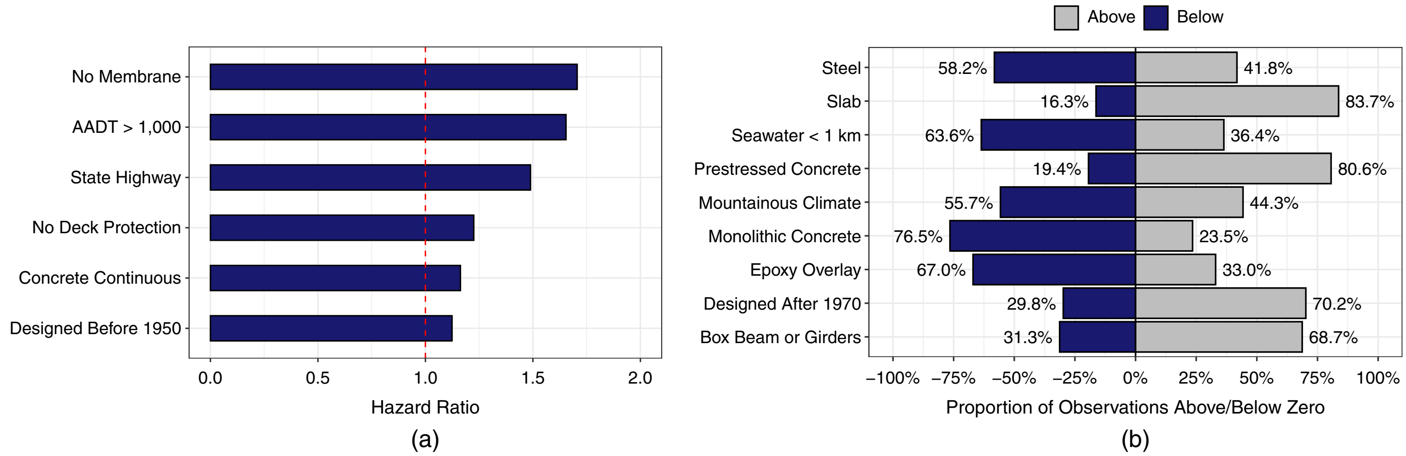

Fig. 1. (a) Estimated hazard ratios of variables in high-CR model; and (b) variables with random parameters and proportion of bridge decks with estimated parameters above and below zero (values above zero indicate a decrease in the likelihood of being assigned a lower CR at any given time, and values below zero indicate an increase in the likelihood of being assigned a lower CR at any given time).

Table 6. Model specifications for medium CR (6) random-parameters survival model

Variable	Coefficient	Standard error	*t*-statistic	Proportional hazards	Hazard ratio
Constant	3.559	0.330	10.79	−6.348	0.002
Maintenance responsibility					
1 if state highway (state responsible), 0 otherwise	0.704	0.155	4.53	−1.256	0.285
1 if county highway (county responsible), 0 otherwise	0.839	0.164	5.11	−1.497	0.224
Kind of material design					
1 if concrete continuous, 0 otherwise	−0.249	0.068	−3.68	0.443	1.558
(Standard deviation, normally distributed)	(0.500)	(0.041)	(12.30)	—	—
1 if steel, 0 otherwise	−0.139	0.083	−1.67	0.248	1.281
(Standard deviation, normally distributed)	(0.810)	(0.060)	(13.39)	—	—
1 if steel continuous, 0 otherwise	−0.333	0.132	−2.51	0.593	1.810
(Standard deviation, normally distributed)	(0.529)	(0.118)	(4.50)	—	—
Deck structure type					
1 if cast-in-place, 0 otherwise	−0.998	0.293	−3.41	1.780	5.931
Type of design construction					
1 if stringer/multibeam girder, 0 otherwise	−0.100	0.052	−1.91	0.178	1.194
Type of wearing surface					
1 if no wearing surface, 0 otherwise	−0.577	0.094	−6.12	1.029	2.800
1 if epoxy overlay, 0 otherwise	−0.080	0.080	−0.99	0.142	1.153
(Standard deviation, normally distributed)	(0.418)	(0.071)	(5.91)	—	—
1 if bituminous, 0 otherwise	0.914	0.092	9.93	−1.631	0.196
(Standard deviation, normally distributed)	(0.969)	(0.065)	(14.87)	—	—
Average daily truck traffic					
1 if greater than 1,000, 0 otherwise	−0.284	0.057	−4.99	0.507	1.660
Climate group					
1 if Climate group 2, 0 otherwise	−0.298	0.065	−4.56	0.531	1.700
(Standard deviation, normally distributed)	(0.416)	(0.055)	(7.54)	—	—
Design period					
1 if after 1970, 0 otherwise	0.398	0.072	5.55	−0.709	0.492
(Standard deviation, normally distributed)	(0.610)	(0.056)	(10.95)	—	—
Scale parameter					
σ	0.561	0.021	26.12	—	—
Model summary					
Number of observations	2,699	—	—	—	—
Number of observations exiting (uncensored)	498	—	—	—	—
Number of observations censored	2,201	—	—	—	—
Log-likelihood at zero	−1,493.11	—	—	—	—
Log-likelihood at convergence	−1,365.96	—	—	—	—
McFadden pseudo R^2	0.09	—	—	—	—

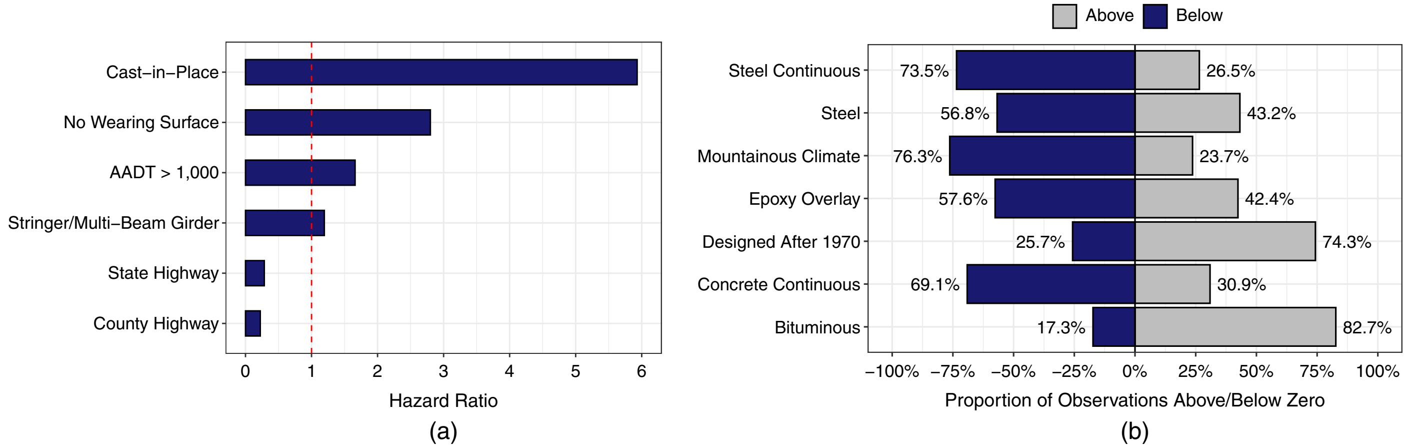

Fig. 2. (a) Estimated hazard ratios of variables in medium-CR model; and (b) variables with random parameters and proportion of bridge decks with estimated parameters above and below zero (values above zero indicate a decrease in the likelihood of being assigned a lower CR at any given time, and values below zero indicate an increase in the likelihood of being assigned a lower CR at any given time).

Table 7. Model specifications for low CR (5 and 6) random-parameters survival model

Variable	Coefficient	Standard error	*t*-statistic	Proportional hazards	Hazard ratio
Constant	3.990	0.159	25.10	−8.378	0.000
(Standard deviation, normally distributed)	(1.080)	(0.077)	(13.95)	—	—
Maintenance responsibility					
1 if county highway (county responsible), 0 otherwise	−0.378	0.138	−2.74	0.794	2.213
Type of design construction					
1 if through truss, 0 otherwise	−0.505	0.172	−2.94	1.061	2.890
Type of wearing surface					
1 if no wearing surface, 0 otherwise	−0.467	0.192	−2.44	0.981	2.668
(Standard deviation, normally distributed)	(0.571)	(0.162)	(3.53)	—	—
1 if latex concrete, 0 otherwise	−0.734	0.134	−5.49	1.542	4.675
Climate group					
1 if Climate group 2, 0 otherwise	−0.338	0.118	−2.88	0.711	2.035
Design period					
1 if after 1970, 0 otherwise	1.409	0.293	4.81	−2.958	0.052
(Standard deviation, normally distributed)	(1.145)	(0.199)	(5.74)	—	—
Scale parameter					
σ	0.476	0.036	13.08	—	—
Model summary					
Number of observations	1,045	—	—	—	—
Number of observations exiting (uncensored)	101	—	—	—	—
Number of observations censored	944	—	—	—	—
Log-likelihood at zero	−373.37	—	—	—	—
Log-likelihood at convergence	−350.75	—	—	—	—
McFadden pseudo R^2	0.06	—	—	—	—

high deterioration in the early years of a bridge deck, but begins to slow as the bridge deck ages.

Box beams or girders also had a normally distributed random parameter. With a mean of 0.266 and a standard deviation of 0.547, 31.3% of high-CR bridge decks with box beam or girder design are more likely to be assigned a lower CR at any given time, and 68.7% are less likely to be assigned a lower CR. One potential reason for these heterogeneous effects may be linked to transverse post-tensioning (TPT) of side-by-side box beam girders. In particular, such elements can begin to show longtidunal cracking over the shear keys within just a few years of initial construction (Grace et al. 2012), thus leading to early deterioration. Grace et al. (2012) showed that TPT can mitigate crack initiation, therefore mitigating deterioration. Another plausible explanation for varying effects may be a result of construction defects, variation in traffic loads, and inaccurate design. Regarding design, shear lag effect in box girders with thin walls, in addition to a potentially intricate prestressing tendon layout, can result in structural design difficulties that lead to accelerated deterioration (Guo et al. 2011; Luo et al. 2002). Regarding variation due to traffic loads, different rates of deterioration can occur due to variations in cyclic loading, volume of truck traffic, and moments generated by unequal wheel loading

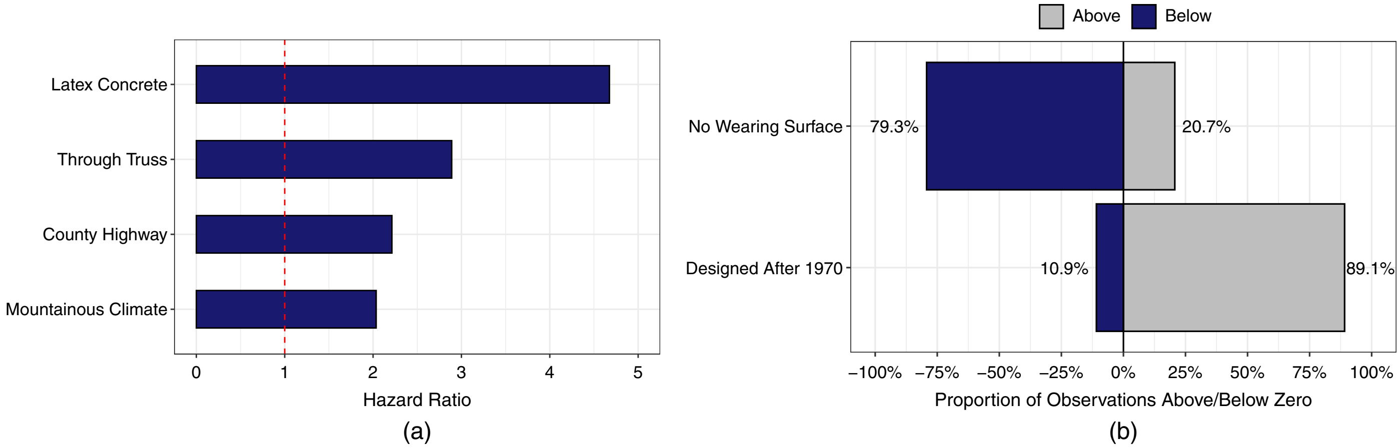

Fig. 3. (a) Estimated hazard ratios of variables in low-CR model; and (b) variables with random parameters and proportion of bridge decks with estimated parameters above and below zero (values nabove zero indicate a decrease in the likelihood of being assigned a lower CR at any given time, and values below zero indicate an increase in the likelihood of being assigned a lower CR at any given time).

Table 8. Chi-squared statistics and degrees of freedom for CR transferability test

	CR_j		
CR_i	High CR	Medium CR	Low CR
High CR	0	3,317.93 (13)	2,339.04 (6)
Medium CR	521.11 (16)	0	700.05 (6)
Low CR	69.01 (16)	123.09 (14)	0

(Sasaki et al. 2010). Specifically, as it pertains to truck traffic, research has shown that overweight trucks substantially increase deterioration over the first several years of a box beam's life, and also in the later years (Lou et al. 2017). This creates heterogeneous effects that are contingent on the age of the box beams. Furthermore, older bridges that were constructed in the early years of noncomposite adjacent box beam construction have increased deterioration due to beam damage and damage to the lateral tie system (e.g., longitudinal cracking of bottom flange, exposure, corrosion, spalling, leakage between beams, and so forth) (Naito et al. 2010), whereas newer bridges would not be subjected to this. Exposure to water also may contribute to the heterogeneous nature of this variable (Naito et al. 2011; Guo et al. 2011).

NBI Item 108A–Type of Wearing Surface

The next two normally distributed random parameters were associated with the type of wearing surface on the bridge deck. The first of these was monolithic concrete, with a mean of −0.260 and a standard deviation of 0.360. These estimates indicate that 23.5% of high-CR bridge decks with a monolithic concrete wearing surface are less likely to be assigned a lower CR at any given time, and 76.5% are more likely to be assigned a lower CR at any given time. Heterogeneous effects on monolithic concrete bridge decks may be due to cracking as a result of varying concrete slump, the percentage volume of water and cement, water content, and decreasing values of air content (Schmitt and Darwin 1995). Additionally, high maximum temperatures and changes in air temperature on the day of casting can contribute to cracking of such decks (Schmitt and Darwin 1995). Another potential explanation for varying effects may be linked to with the durability of the concrete used for construction (De Leon 2018).

The second wearing surface with a normally distributed random parameter was epoxy overlay. The model estimated a mean of −0.223 and a standard deviation of 0.508. These estimates indicate that 33.0% of high-CR bridge decks with an epoxy overlay wearing surface are less likely to be assigned a lower CR at any given time, and 67.0% are more likely to be assigned a lower CR at any given time. The finding that the majority of bridge decks experience high deterioration agrees with previous work, in which epoxy overlay was found to increase deterioration considerably (Goyal 2015; Manafpour et al. 2018). In regard to variation in deterioration rates, plausible reasons may include poor construction and inadequate deck preparation (which can lead to delamination and result in increased deterioration). Additionally, the use of soft aggregates may result in increased wear due to traffic and/or snow equipment (Guthrie et al. 2005; DeRuyver and Scheifer 2016; De Leon 2018).

The final three random parameters represented three different variable categories.

Distance to Seawater

The first of these variables with a normally distributed random parameter was the distance to seawater. With a mean of −0.323 and a standard deviation of 0.928, 36.4% of high-CR bridge decks within 1 km from seawater are less likely at any given time, and 63.6% are more likely to be assigned a lower CR. A possible explanation for these varying effects may be related to the use of special sulfate-resistant concrete (Gucunski et al. 2011). Height above seawater also could play a role, but some work has shown that above a specific height, the effect of height above seawater chloride concentration does not change significantly (Vu and Stewart 2000). Varying effects also may be attributed to the fact that some coast bridges have cathodic protection, whereas others may not.

Climate Group 2: Cascade Mountain Range

The next variable with a normally distributed random parameter was the indicator for Climate group 2 (climate zones along the Cascade Mountain Range). With a mean of −0.102 and a standard deviation of 0.711, 44.3% of high-CR bridge decks in Climate group 2 are less likely to be assigned a lower CR at any given time, and 55.7% are more likely to be assigned a lower CR at any given time. The variation observed here may be attributed to within-climate group variations in precipitation and freeze–thaw cycles.

Design Period

The final variable with a normally distributed estimated random parameter was the indicator for the design period after 1970. A

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mean of 0.196 and a standard deviation of 0.369 indicated that 29.8% of high-CR bridge decks that were designed after 1970 are more likely to be assigned a lower CR at any given time, and 70.2% are less likely to be assigned a lower CR at any given time.

Large-Effect Variables in High-CR Model

Of the variables with high effects on the likelihood of a bridge deck being assigned a lower CR at any given time, four were notable, all of which increase the likelihood of being assigned a lower CR at any given time. The variable indicated that bridge decks that fall under the maintenance responsibility of state highway (state responsible) are 48.9% more likely to be assigned a lower CR at any given time. Bridge decks that have no membrane are 70.9% more likely to be assigned a lower CR at any given time. Bridge decks with no protection are 22.5% more likely to be assigned a lower CR at any given time. Lastly, bridge decks that experience average daily truck traffic greater than 1,000 are 65.4% more likely to be assigned a lower CR at any given time.

Medium-CR Group

For the medium-CR model, 13 variables were found to be significant contributing factors to TICR. Of these 13 variables, 7 had heterogeneous effects on a bridge deck's TICR.

NBI Item 43A–Structure Type: Kind of Material Design

Three of the seven variables with heterogeneous effects were associated with the kind of material design. First, the indicator for concrete continuous material design had a normally distributed random parameter with a mean of −0.249 and a standard deviation of 0.500. These estimates indicate that 30.9% of concrete continuous medium CR bridge decks are less likely to be assigned a lower CR at any given time, and 69.1% are more likely to be assigned a lower CR. Heterogeneous effects of concrete continuous on the probability of high bridge deck deterioration also were found by Ghonima et al. (2020). As suggested by Ghonima et al. (2020), the heterogeneous effects may be attributed to unobservable factors related to cracking (e.g., water and chemicals penetrating the deck) (Schmitt and Darwin 1995). Negative bending momements at regions of interior supports have been shown to result in such cracking (Grace et al. 2004; Manafpour et al. 2018; Fleischhacker et al. 2020). Ghonima et al. (2020) suggested that these heterogeneous effects can be linked to vibration severity, which can be bridge-deck specific (Alampalli et al. 2002).

Secondly, the indicator for steel material design had a normally distributed random parameter. With a mean of −0.139 and a standard deviation of 0.810, 43.2% of steel medium CR bridge decks are less likely to be assigned a lower CR at any given time, and 56.8% are more likely to be assigned a lower CR at any given. Third, the indicator for steel continuous material design had a normally distributed random parameter. With an estimated mean of −0.333 and an estimated standard deviation of 0.529, 26.5% of steel continuous medium CR bridge decks are less likely to be assigned a lower CR at any given time, and 73.5% is more likely to be assigned a lower CR at any given time. As discussed for the high-CR group, the variation due to steel material design can be a result of various factors.

NBI Item 108A–Type of Wearing Surface

Two random parameters were associated with the type of wearing surface on the bridge deck. Specifically, bridge decks with an epoxy overlay wearing surface were found to have a normally distributed random parameter with a mean of −0.080 and a standard deviation of 0.418. These estimates suggest that 42.4% of medium CR bridge decks with an epoxy overlay wearing surface are less likely to be assigned a lower CR at any given time, whereas 57.6% of medium CR bridge decks with an epoxy overlay wearing surface are more likely to be assigned a lower CR at any given time. Referring to the discussion of epoxy overlay in the high-CR model, possible explanations for bridge-deck-specific variation may be linked to poor construction, inadequate deck preparation, and the use of soft aggregates, as discussed for steel material design for the high-CR model.

The second wearing surface indicator found to have a random parameter was bituminous. With a mean of 0.914 and a standard deviation of 0.969, 17.3% of medium CR bridge decks with a bituminous wearing surface are more likely to be assigned a lower CR at any given time, and 82.7% of such bridge decks are less likely to be assigned a lower CR at any given time. A potential reason for heterogeneous effects observed here may be linked to early or late treatment of a bituminous overlay (De Leon 2018). De Leon (2018) found that early treatment can extend bridge deck service life from 8 years to more than 10 years, whereas late treatment can extend bridge deck service life by more than 10 years. Another potential reason for these varying effects could be attributed to waterproofing (De Leon 2018; Mazzotta et al. 2017; Kosior-Kazberuk and Jezierski 2004). It also has been found that a bituminous wearing surface can fail early as a result of inadequate mixture design and/or poor construction, which can vary from deck to deck and result in deck-specific longitudinal cracking and transverse cracking (Battaglia and Peters 2012).

Climate Group 2: Cascade Mountain Range

Of the final two random parameters in the medium-CR model, the first was related to climate. In particular, with a mean of −0.298 and a standard deviation of 0.416, 23.7% of medium CR bridge decks located in Climate group 2 are less likely to be assigned a lower CR at any given time, and 76.3% of medium CR bridge decks located in Climate group 2 are more likely to be assigned a lower CR.

Design Period

The final random parameter in the medium-CR model was the indicator for bridges that were designed after 1970. With a mean of 0.398 and a standard deviation of 0.610, 25.7% of medium CR bridge decks designed after 1970 are more likely to be assigned a lower CR at any given time, and 74.3% of medium CR bridge decks designed after 1970 are less likely to be assigned a lower CR.

Large-Effect Variables in Medium-CR Model

Of the variables with large effects on the likelihood of a bridge deck being assigned a lower CR at any given time, two decreased the likelihood, and three increased the likelihood. The two variables associated with a decrease were maintenance responsibility indicators; specifically, state highway (state responsible) and county highway (county responsible). According to estimated hazard ratios, bridge decks that fall under the maintenance responsibility of state highway (state responsible) are nearly 70% less likely to be assigned a lower CR at any given time, and bridge decks under the maintenance responsibility of county highway (county responsible) are nearly 80% less likely to be assigned a lower CR at any given time. Of the variables that increased the likelihood of being assigned a lower CR any given time, cast-in-place deck structure type had the largest effect; the hazard ratio showed that cast-in-place deck structure type was nearly six times more likely to be assigned a lower CR at any given time than were other deck structure types. Another variable with a moderate impact was no wearing surface—bridge decks with no wearing surface are nearly three times more likely to be assigned a lower CR at any given time than are other wearing surface types. The final variable with a moderate impact was the indicator for average daily truck traffic greater than

1,000—bridge decks that experience such truck traffic are 66% more likely to be assigned a lower CR at any given time.

Low-CR Group

The final model was for low CR bridge decks (CRs of 4 and 5). Unlike the previous two CR groups, only six variables were found to be significant, and two had heterogeneous effects (the constant also was found to be random).

NBI Item 108A–Type of Wearing Surface

Of the two variables with heterogeneous effects on TICR, the first was related to the type of wearing surface. With a mean of −0.467 and a standard deviation of 0.571, 20.7% of low CR bridge decks with no wearing surface are less likely to be assigned a lower CR at any given time, and 79.3% of low CR bridge decks with no wearing surface are more likely to be assigned a lower CR at any given time. Heterogeneous effects due to no wearing surface also were found by Ghonima et al. (2020). Specifically, Ghonima et al. (2020) linked varying effects due to no wearing surface with limit states and the use or nonuse of deicers.

Design Period

The second variable with heterogeneous effects was the indicator for bridges designed after 1970. With an estimated mean of 1.409 and an estimated standard deviation of 1.145, 10.9% of low CR bridge decks designed after 1970 are more likely to be assigned a lower CR at any given time, and 89.1% of low CR bridge decks designed after 1970 are less likely to be assigned a lower CR at any given time.

Large-Effect Variables in Low-CR Model

In terms of effects on the likelihood of being assigned a lower CR at any given time, the effects in the low-CR model in general, were much greater than in the other CR models. Pointedly, bridge decks that fall under the responsibility of county highway (county responsible) are more than 2 times more likely to be assigned a lower CR at any given time than are bridge decks that fall under the maintenance responsibility of a different jurisdiction. The indicator for through truss type of design construction resulted in bridge decks that are nearly 3 times more likely to be assigned a lower CR at any given time compared to other types of design construction. The indicator for latex concrete wearing surface also had substantial effects on the likelihood of being assigned a lower CR at any given time; specifically, bridge decks with latex concrete wearing surfaces are more than 4 times as likely to be assigned a lower CR at any given time than are bridge decks with other wearing surfaces. Finally, bridge decks in Climate group 2 are more than two times as likely to be assigned a lower CR at any given time than are bridge decks in other climate groups.

Comparison Across CR Models

In comparing variables and effects across the three CR models, variables found to be significant in at least two of the CR groups were of particular interest. Therefore, comparisons were made based on this premise. In addition, Table 9 comprehensively summarizes the different variables and their effects across all three CR models.

Factors Significant in Two of Three CR Models

A total of six variables were found to be significant in two of the three CR models. Of the six variables, five were significant among the high-CR model and medium-CR model. The first of these variables was related to the kind of material design: concrete continuous. In the high-CR model, concrete continuous had homogeneous effects across bridge decks, increasing TICR. In the medium-CR model, concrete continuous was found to have heterogeneous effects across bridge decks. Table 9 lists the percentage of observations with an increase in TICR and the percentage of observations with a decrease in TICR. The other variable associated with material design significant in both the high-CR and medium-CR models was the indicator for steel. In both models, this indicator was found to be heterogeneous across bridge decks. Fig. 4 shows how the $\boldsymbol{\beta}$ estimates varied across observations based on a normal distribution. The heterogeneous effects of steel in both CR models were nearly the same: 58.2% of bridge decks in the high-CR model are more likely to be assigned a lower CR at any given time, compared with 56.8% of bridge decks in the medium-CR model. Likewise, 41.8% of bridge decks in the high-CR model are less likely to be assigned a lower CR at any given time, compared with 43.2% of bridge decks in the medium-CR model.

Another variable that was significant in the high-CR and medium-CR models, and with heterogeneous effects, was bridge decks with an epoxy overlay wearing surface. Fig. 5 shows how $\boldsymbol{\beta}$ estimates varied across observations based on a normal distribution. Unlike steel material design (Fig. 4), the heterogeneous effects were moderately different by CR group (Fig. 5). In particular, 33.0% of bridge decks with an epoxy overlay wearing surface in the high-CR model are less likely to be assigned a lower CR at any given time, compared with 42.4% in the medium-CR model. Similarly, 67.0% of bridge decks with an epoxy overlay wearing surface in the high-CR model are more likely to be assigned a lower CR at any given time, compared with 57.6% in the medium-CR model.

Of the last two variables that were significant in both the high-CR and medium-CR models, one was associated with maintenance responsibility, and one was related to average daily truck traffic. In regard to maintenance responsibility, the effects differed by CR group. In the high-CR model, bridge decks that fall under the maintenance responsibility of state highway (state responsible) are approximately 49% more likely to be assigned a lower CR at any given time. On the other hand, in the medium-CR model, bridge decks that fall under the maintenance responsibility of state highway (state responsible) are just over 70% less likely to be assigned a lower CR at any given time. The final variable significant between the two models was the indicator for maintenance responsibility that falls under county highway (county responsible), which was shared among the medium CR and the low-CR model. The effects of this indicator differed substantially by CR model. In the medium-CR model, bridge decks that fall under the maintenance responsibility of county highway (county responsible) are over 75% less likely to be assigned a lower CR at any given time than are bridge decks under the maintenance responsibility of other jurisdictions. This variable had a quite different impact in the lower-CR model: bridge decks that fall under the maintenance responsibility of county highway (county responsible) are more than 2 times as likely to be assigned a lower CR at any given time than are bridge decks under the maintenance responsibility of other jurisdictions.

Factors Significant in All Three CR Models

Two variables were significant in all three CR models: the indicator for Climate group 2 and the indicator for the design period after 1970. on the effects of the climate indicator were heterogeneous in the high- and medium-CR models, whereas the effects were homogeneous in the low-CR model. Fig. 6 illustrates how $\boldsymbol{\beta}$ estimates varied under the normal distribution comparison with the homogeneous effects in the low-CR model. The heterogeneous effects in the high-CR model and medium-CR model were substantially different; 44.3% of bridges in the high-CR group are less

Table 9. Significant variables and effects on TICR among CR models

Variable	High-CR model	Medium-CR model	Low-CR model
Maintenance responsibility			
1 if state highway (state responsible), 0 otherwise	+ (1.489)	− (0.285)	—
1 if county highway (county responsible), 0 otherwise	—	− (0.224)	+ (2.213)
Kind of material design			
1 if concrete continuous, 0 otherwise	+ (1.163)	+ − (69.1%, 30.9%)	
1 if steel, 0 otherwise	+ − (58.2%, 41.8%)	+ − (56.8%, 43.2%)	—
1 if steel continuous, 0 otherwise	—	+ − (73.5%, 26.5%)	—
1 if prestressed concrete, 0 otherwise	+ − (19.4%, 80.6%)	—	—
Deck structure type			
1 if cast-in-place, 0 otherwise	—	+ (5.931)	—
Type of design construction			
1 if slab, 0 otherwise	+ − (16.3%, 83.7%)	—	—
1 if box beam or girders, 0 otherwise	+ − (31.3%, 68.7%)	—	—
1 if stringer/multibeam girder, 0 otherwise	—	+ (1.194)	—
1 if through truss, 0 otherwise	—	—	+ (2.890)
Type of wearing surface			
1 if no wearing surface, 0 otherwise	—	+ (2.800)	+ − (79.3%, 20.7%)
1 if monolithic concrete, 0 otherwise	+ − (76.5%, 23.5%)	—	—
1 if epoxy overlay, 0 otherwise	+ − (67.0%, 33.0%)	+ − (57.6%, 42.4%)	—
1 if bituminous, 0 otherwise	—	+ − (17.3%, 82.7%)	—
1 if latex concrete, 0 otherwise	—	—	+ (4.675)
Type of membrane			
1 if no membrane, 0 otherwise	+ (1.706)	—	—
Deck protection			
1 if no deck protection, 0 otherwise	+ (1.225)	—	—
Average daily truck traffic			
1 if greater than 1,000, 0 otherwise	+ (1.654)	+ (1.660)	—
Distance to seawater			
1 if less than or equal to 1 km, 0 otherwise	+ − (63.6%, 36.4%)	—	—
Climate group			
1 if Climate group 2, 0 otherwise	+ − (55.7%, 44.3%)	+ − (76.3%, 23.7%)	+ (2.035)
Design period			
1 if before 1950, 0 otherwise	+ (1.124)	—	—
1 if after 1970, 0 otherwise	+ − (29.8%, 70.2%)	+ − (25.7%, 74.3%)	+ − (10.9%, 89.1%)

Note: + = increased likelihood of being assigned a lower CR at any given time (estimated hazard ratio); − = decreased likelihood of being assigned a lower CR at any given time (estimated hazard ratio); + − = heterogeneous (% of observations more likely to be assigned a lower CR at any given time, % of observations less likely to be assigned a lower CR at any given time); and — = insignificant in corresponding CR model.

likely to be assigned a lower CR at any given, compared with 23.7% in the medium-CR group. Conversely, 55.7% of bridge decks in the high-CR group are more likely to be assigned a lower CR at any given time, compared with 76.3% in the medium-CR group.

The second variable that was significant among all three models was the indicator for the design period after 1970. This indicator had heterogeneous effects in each model. Fig. 7 illustrates how the $\boldsymbol{\beta}$ estimates varied by CR model. In each model, the majority of bridge decks in which the design period was after 1970 are less likely to be assigned a lower CR at any given time. Specifically, 29.8% in the high-CR model, 25.7% in the medium-CR model, and 10.9% in the low-CR model are more likely to be assigned a lower CR at any given time. Conversely, 70.2% in the high-CR model, 74.3% in the medium-CR model, and 89.1% in the low-CR model are less likely to be assigned a lower CR at any given time.

Summary

Using NBI data, along with key Oregon-specific variables (climate and design period), a series of log-linear Weibull survival models was fit for three different CR groups. The use of survival models accounts for the substantial amount of censorship observed in the data, ranging from 77% in the high-CR group to more than 90% in the low-CR group. Using this framework, CR-group-specific effects of NBI-related variables on TICR were determined. This expressly differed from previous work in which CR groups often were used as covariates in a single, holistic model. Additionally, to account for data heterogeneity, parametric survival models were fit. Using the log-linear Weibull survival model, unobserved heterogeneity was addressed through the estimation of random parameters. Based on the number of random parameters in two of the three CR models, the NBI data are susceptible to large amounts of heterogeneity (nine random parameters in the high-CR model and seven random parameters in the medium-CR model). Lastly, a parameter transferability test was conducted to determine if using three separate survival models based on CR group was appropriate. Results from the parameter transferability test indicated with well over 99% confidence that TICR should be analyzed independently by CR group. That is, the null hypothesis that parameter estimates are transferable between CR groups was rejected.

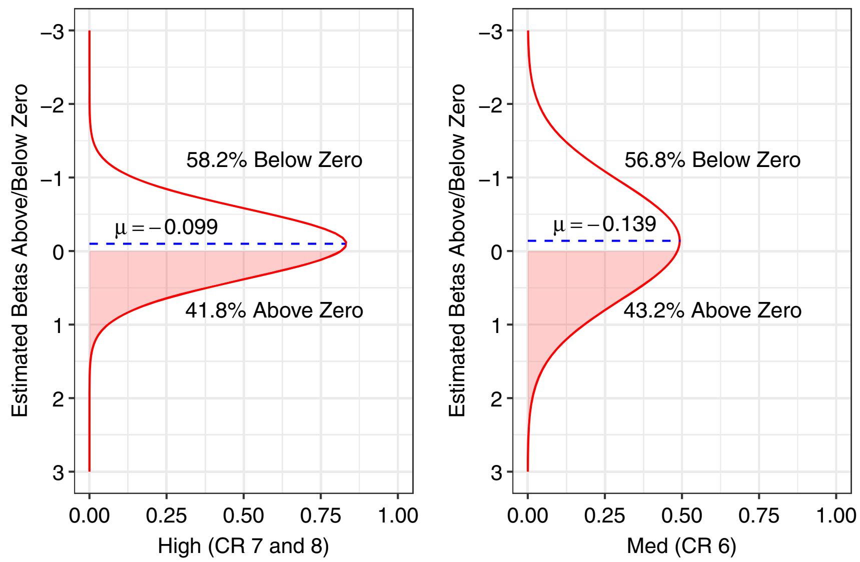

Fig. 4. Comparison of heterogeneous effects of steel material design on TICR in high-CR and medium-CR models (positive β values indicate a decrease in the likelihood of being assigned a lower CR at any given time, and negative β values indicate an increase in the likelihood of being assigned a lower CR at any given time).

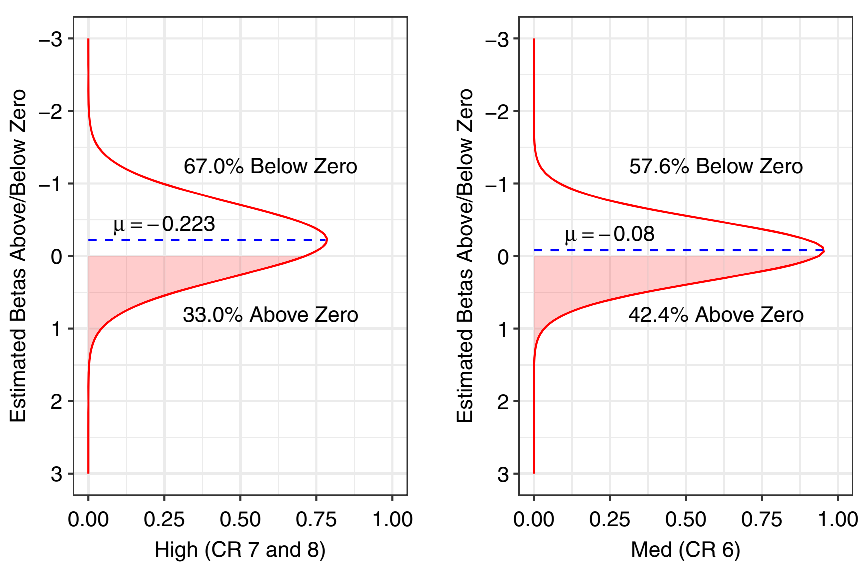

Fig. 5. Comparison of heterogeneous effects of epoxy overlay wearing surface on TICR in high-CR and medium-CR models (positive β values indicate a decrease in the likelihood of being assigned a lower CR at any given time, and negative β values indicate an increase in the likelihood of being assigned a lower CR at any given time).

In the high-CR model, the indicator for no membrane had the largest impact on TICR—bridge decks in the high-CR group with no membrane are approximately 70% more likely to be assigned a lower CR at any given time. In the medium-CR model, the largest impact on TICR was the indicator for cast-in-place deck structure type—the cast-in-place deck structure type is nearly 6 times more likely to be assigned a lower CR at any given time than are other deck structure types (this also was the largest effect across all three CR models). In the low-CR model, the indicator for latex concrete wearing surface had the largest effects on TICR, in which latex concrete wearing surfaces are more than 4 times more likely to be assigned a lower CR at any given time than are other wearing surfaces. Overall, excluding cast-in-place in the medium-CR model, the largest effects on TICR were observed in the low-CR model. These results suggest that a bridge deck with a lower (CR 4 and CR 5) has a substantially higher likelihood of being assigned a lower CR at any given time than bridge decks with higher CRs.

Of the variables that were significant in two of the CR models, steel material design was heterogeneous in the high- and medium-CR models—the proportion of negative and positive β values was approximately the same in both groups; there was a slight majority of positive values, indicating a decrease in the likelihood of being assigned a lower CR at any given time. Epoxy overlays also were found to have heterogeneous effects in the high- and medium-CR models. The differences in β distributions were more profound—the proportions were approximately 10% higher/lower in terms of increasing and decreasing likelihoods of being assigned a lower CR at any given time.

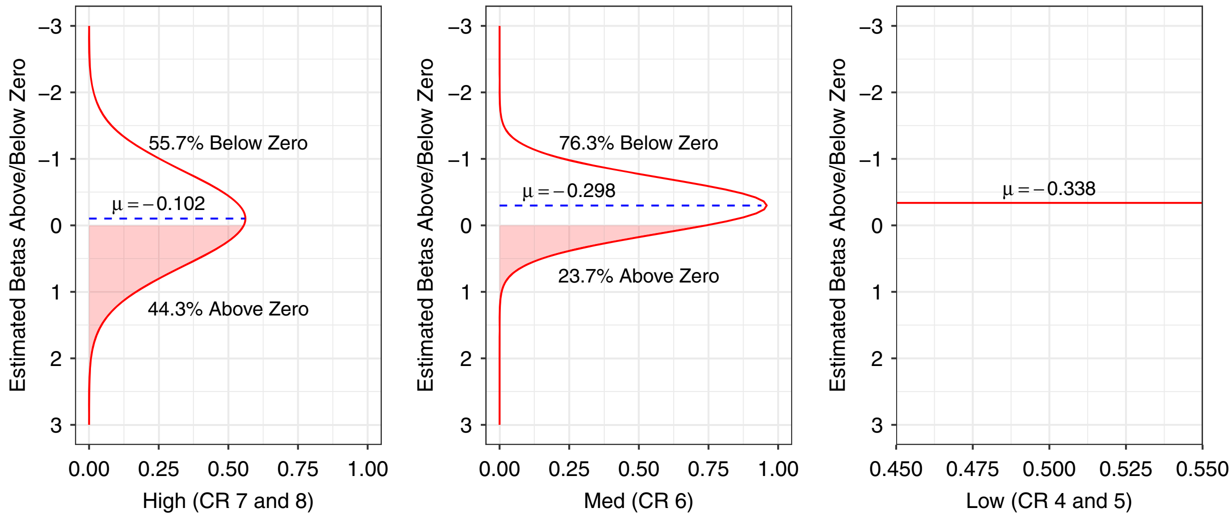

Fig. 6. Comparison of heterogeneous effects of Climate group 2 on TICR in high-CR, medium-CR, and low-CR models (positive β values indicate a decrease in the likelihood of being assigned a lower CR at any given time, and negative β values indicate an increase in the likelihood of being assigned a lower CR at any given time).

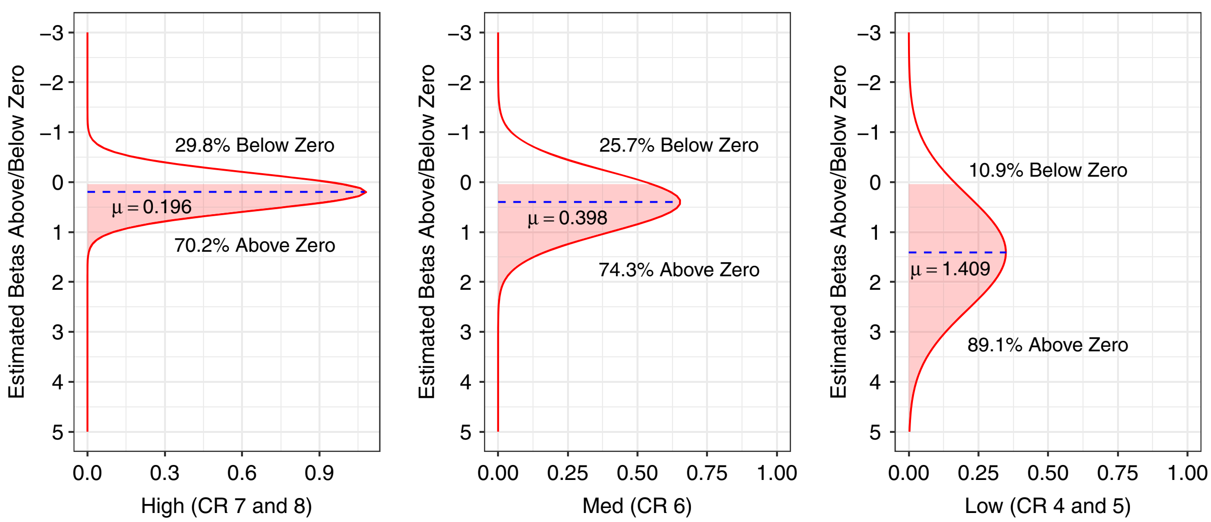

Fig. 7. Comparison of heterogeneous effects of design period after 1970 on TICR in high-CR, medium-CR, and low-CR models (positive β values indicate a decrease in the likelihood of being assigned a lower CR at any given time, and negative β values indicate an increase in the likelihood of being assigned a lower CR at any given time).

Two variables were found to be significant in all three models. The indicator for Climate group 2 (Cascade Mountain Range) was heterogeneous in the high- and medium-CR models, and homogeneous in the low-CR model. In the high-CR model, 55.7% of bridge decks are more likely to be assigned a lower CR at any given time; in the medium-CR model, 76.3% were more likely are more likely to be assigned a lower CR at any given time; and in the low-CR model, all bridge decks are more likely (more than 2 times as likely compared with other climate groups). These results suggest that as CR decreases, bridge decks in mountainous climates are increasingly likely to be assigned a lower CR at any given time. Design period after 1970 also was found to be significant in all three models, and had heterogeneous effects in all three CR models. In the high- and medium-CR models, the effects were similar; specifically, 29.8% are more likely to be assigned a lower CR at any given time in the high-CR model compared with 25.7% in the medium-CR model. However, in the lower-CR model, this decreases to 10.9%. These results suggest that the impact of being designed after 1970 is more profound for lower CR bridge decks.

Although this study addressed explicit gaps in the literature by focusing on CR-group-specific effects and unobserved heterogeneity through random parameter estimation, there are avenues for future research. For example, future work should address both of the aforementioned components, as well as interval-censorship. Furthermore, although this work addressed bridge-deck-specific variation, heterogeneity in the means of the random parameters was not tested, nor was correlation between random parameters. On the premise of prediction, this work focused explicitly on variable impacts by CR group, not prediction-based metrics. This provides an avenue for future research on prediction and CR groups; for example, predictions on high-CR groups, medium-CR groups, low-CR groups, and how these compare to a traditional method that encompasses all CRs when assessing prediction performance. This work showed that CR groups should be analyzed separately, indicating that future work considering different modeling frameworks should disaggregate the NBI data by CR group. Lastly, future work that uses NBI data should consider separate analyses by data subsets of interest.

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request—specifically, the NBI data set for Oregon.

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